

NATARAJAN BALAJI SHANKAR

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Education

University of California, Los Angeles (UCLA)

July 2023 – Expected Jun 2027

Doctor of Philosophy, Electrical and Computer Engineering

Los Angeles, CA

- Advanced to Ph.D. candidacy December 2024; cumulative GPA 3.975.

University of California, Los Angeles (UCLA)

Sep 2021 – Jun 2023

Master of Science, Electrical and Computer Engineering

Los Angeles, CA

National Institute of Technology, Tiruchirappalli (NIT Trichy)

July 2016 – Jun 2020

Bachelor of Technology, Electronics and Communication Engineering

Tiruchirappalli, India

- First Class with Distinction; Minor in Computer Science and Engineering.

Technical Skills

Languages: Python, C, C++, MATLAB, Bash, SQL

Speech & ASR Toolkits: ESPnet, SpeechBrain, fairseq, NeMo, Kaldi, k2, torchaudio, librosa, FFmpeg

Deep Learning & ML: PyTorch, HF Transformers, NumPy, pandas, SentencePiece

LLM & GenAI: RAG, QLoRA, bitsandbytes, Ollama, LLM Agent Orchestration, Agentic Coding (Claude Code, Codex)

Infrastructure & Tools: Linux, Docker, Git, AWS, SLURM, Distributed Training (DDP, DeepSpeed, FSDP), Hydra, Weights & Biases

Research Areas: Automatic Speech Recognition, Self-Supervised Learning, Speech Foundation Models, Domain Adaptation, Parameter-Efficient Finetuning (PEFT), Neural Audio Coding, On-Device & Efficient Speech Processing, Speaker Representation Learning, Speech-LLMs

Experience

Speech Processing and Auditory Perception Laboratory

01/2022 - Present

Graduate Student Researcher

Los Angeles, CA

- Research supported by NSF Award #2202585 and U.S. Department of Education IES Grant #R305C240046.

Model Merging for Low-Resource Automatic Speech Recognition

- Developed Selective Attention (SA) Merge and a headwise variant (HSA Merge), a novel approach that merges task vectors from attention matrices, enhancing Speech Foundation Model (SFM) performance for low-resource tasks such as child Automatic Speech Recognition.
- Achieved up to a 20.6% and 29.2% relative Word Error Rate (WER) reduction on child ASR and cross-dialect transfer by engineering a structured, attention-level parameter merging technique for the Whisper model family.
- Extended the framework to compositional domain adaptation, showing that adaptations for distinct distribution shifts (acoustics, speaking style, dialect) can be recombined to generalize to unseen domain intersections without retraining.

Child ASR Hypothesis Error Correction

- Conducted comprehensive analyses of transcription errors made by Automatic Speech Recognition (ASR) models on child speech, identifying error patterns linked to acoustic and linguistic variations.
- Curated CHSER, a dataset of paired ASR hypotheses and references for child speech, and investigated correction frameworks leveraging Large Language Models (LLMs) and speech-LLMs, developing approaches for acoustically conditioned child speech error correction.

Hallucination Mitigation in Speech Foundation Models

- Developed AURA, an ultra-efficient activation-editing method that keeps Whisper frozen and learns sparse scale-and-shift edits on individual decoder cross-attention heads, with a token-level gate that routes edit strength using cross-attention uncertainty (over-concentration, diffuse attention, and abrupt attention shifts).
- Sharply reduced Whisper-large-v3 hallucinations on non-speech audio without identifying hallucination heads beforehand, and approached LoRA WER on child, imperfectly labeled, and disfluent speech with a small fraction of LoRA’s trainable parameters.

Improving Child Automatic Speech Recognition

- Introduced novel Speech-Only Adaptation (SOA) technique for unsupervised domain adaptation of speech foundation models to low-resource domains utilizing only audio data, resulting in a 29% relative WER reduction.
- Proposed Perturbation Invariant Finetuning (PIF) loss to stabilize finetuning on augmented child speech, achieving state-of-the-art results on the MyST dataset.
- Designed GC-LoRA, a Conformer-style convolutional adapter that augments Low-Rank Adaptation with local acoustic context, achieving up to 10.9% relative WER reduction across acoustically degraded, bandlimited, dialectal, and child speech while adding minimal trainable parameters.
- Curated the JIBO Kids Corpus, a speech dataset of child-robot interactions recorded in classroom environments, supporting research on children’s conversational speech and child ASR.
- Benchmarked pretrained (supervised and self-supervised) and trained-from-scratch ASR models across a range of child speech datasets, systematically studying foundation-model choice, training paradigms, dataset composition, and model scaling to establish reproducible child ASR baselines; the resulting benchmark holds the best published WER on the MyST children’s corpus (8.5%).

Automated Child Language Assessment

- Developed framework for automatically scoring children’s oral narrative language abilities using linguistic features and BERT-based representations, achieving 98% classification accuracy on a portion of the Test of Narrative Language.
- Evaluated open-source ASR systems (Whisper, HuBERT, Wav2Vec2) on African American English-speaking children’s speech and used BERT to grade oral responses with 95% accuracy.
- Built an end-to-end ASR + LLM pipeline that returns rubric-aligned scores together with qualitative feedback for children’s spoken language assessments, approaching human-level scoring agreement while remaining interpretable.

Encoder-Only Model for Non-autoregressive Speech Recognition

- Devised UniEnc-CASSNAT, a novel encoder-based non-autoregressive ASR model combining CTC and attention decoding, leveraging a single encoder for mask prediction and text decoding.
- Produced state-of-the-art results on Librispeech, MyST, and Aishell1 datasets with fewer model parameters than existing NASR models.

Question Answering from Long Context Audio

- Created a novel dataset (CORAAL QA) and framework for audio question-answering from long audio recordings of spontaneous speech, utilizing PLDA scoring of semantic embeddings to rank segment relevance for answering questions.
- Employed large language models (ChatGPT, Llama 2) to generate additional training examples, enhancing framework performance through data augmentation.

Predicting Dialect Density in African American English

- Formulated framework for predicting dialect density in African American English speakers integrating HuBERT representations, prosody, grammar, and speaker embedding-based feature sets.
- Enhanced specificity in downstream tasks by predicting the level of dialect density, achieving a 40% accuracy improvement over baselines in adults and over 70% accuracy in children.

Qualcomm Incorporated

Audio & ML Research Intern

06/2025 - 09/2025, 06/2026 - 09/2026

San Diego, CA

On-Device Speaker Normalization for Robust Speech Front-Ends (2026)

- Developed low-complexity, on-device talker normalization pipelines that suppress speaker variability in the acoustic front-end, improving the robustness of downstream automatic speech recognition and voice activity detection (VAD) under real-time compute and memory budgets.
- Designed retrieval-based speaker normalization that adapts speech features on the fly against a stored bank of reference speaker embeddings, adding negligible parameter count and inference overhead relative to speaker-adaptive retraining.

Low-Bitrate Neural Speech Coding (2025)

- Developed and optimized multi-bitrate speech coding methods aimed at enabling practical deployment in resource-constrained conditions.
- Contributed to research on neural speech coding through novel approaches for balancing speech quality and bitrate efficiency in low-resource scenarios.
- Designed and evaluated codec performance under challenging real-world conditions, including noise, packet loss, and speaker variability.
- Named inventor on ten U.S. provisional patent applications covering five inventions in speech technology, filed by Qualcomm with the USPTO in November 2025.

KLA Corporation

Algorithms Intern

06/2022 - 09/2022

Milpitas, CA

- Modernized a wafer inspection tool by migrating its backend to Linux, enabling multi-GPU execution and boosting performance.
- Engineered a data transmission framework for a wafer inspection tool GUI to seamlessly communicate with a Linux-based backend.
- Developed a secure Python/Flask license generation system to authenticate client access for an internal tool.

Patents

- Named inventor on **ten U.S. provisional patent applications** covering five inventions in speech processing technology, filed by Qualcomm Incorporated with the USPTO in November 2025. Application contents confidential pending publication.

Research Impact and Adoption

- The **JIBO Kids Corpus** (Zenodo DOI: 10.5281/zenodo.13964792) is the official training-data source, with required citation, of the Gates Foundation-funded DrivenData Children’s Speech Recognition Challenge (\$120,000 in prizes; 828 participants); additionally used in independent research at the University of Arizona, Vanderbilt University Medical Center, and Université de Moncton.
- Child ASR benchmark adopted as the experimental protocol by independent groups at USC, UT Dallas, CMU, and Stellenbosch University; model merging methods built on or extended by research groups in France, Portugal, and Singapore.
- Released **100 open speech models on Hugging Face** (3,736 lifetime downloads), including child-adapted Whisper, WavLM, and HuBERT model families; trained models and code additionally requested by researchers at CMU, Columbia, Stanford, the University of Washington, and UC San Diego.

Academic and Research Contributions

- Author of **26 peer-reviewed publications** (7 journal, 19 conference; 9 first-author), with **224 citations and an h-index of 8** (Google Scholar, September 2026).
- **Best Paper Award**, AI for Children’s Speech and Language Workshop (AI4CSL), IEEE ASRU 2025 Satellite Workshop, Honolulu, December 2025.
- **2025 Amazon Fellow**, UCLA Science Hub for Humanity and Artificial Intelligence (UCLA–Amazon partnership); 13 doctoral fellows selected university-wide from 47 applicants, recognized for contributions to responsible and low-resource speech and language processing.
- Peer reviewer for **71 manuscripts (42 conference, 29 journal) across 20 international venues**, including IEEE/ACM Transactions on Audio, Speech, and Language Processing; IEEE Signal Processing Letters; Scientific Reports; npj Digital Medicine; Frontiers in Digital Health; Signal, Image and Video Processing; ICASSP 2025 and 2026; Interspeech 2026; IEEE SLT 2026; EUSIPCO 2026; IJCNN 2025–2026; and SLaTE 2025.
- Mentored nine Master’s and undergraduate researchers across child ASR, diarization, and assessment projects, leading to completed theses, course projects, and co-authored publications.
- Taught as an Associate Instructor and Teaching Assistant for nine quarters across five UCLA courses, including EC ENGR 114 (Speech and Image Processing) and ECE M214A (Digital Speech Processing).

Publications

- **Shankar, N. B.**, Wang, Zilai, Wang, Zihan, Shi, M., Zhang, K., & Alwan, A. (2026). AURA: Uncertainty-Routed Activation Editing for Acoustic Grounding in Speech Foundation Models. *IEEE Spoken Language Technology Workshop (SLT) 2026* (to appear).
- Shi, M., Wang, Z., **Shankar, N. B.**, Zhang, K., Eren, E., & Alwan, A. (2026). Encoder Awakening via Adapters: Effective Domain-Adaptive Fine-tuning of Speech-LLMs. *IEEE Spoken Language Technology Workshop (SLT) 2026* (to appear).
- **Shankar, N. B.**, Wang, Z., Zhang, K., Shi, M., & Alwan, A. (2026). GC-LoRA: Gated Convolutional LoRA for Parameter-Efficient Acoustic Adaptation. *Interspeech 2026*
- Wang, Z., **Shankar, N. B.**, Shi, M., Zhang, K., & Alwan, A. (2026). Gumbel-BEARD: Automatic Layer Selection for Self-Supervised Adaptation of Whisper in Low-Resource Domains. *Interspeech 2026*
- Shi, M., Zhang, K., Wang, Z., **Shankar, N. B.**, Eren, E., & Alwan, A. (2026). Entropy-Aware Domain-Routed Mixture-of-Experts Speech-LLM Framework: A Case Study of Multi-Domain Child-Adult ASR. *Interspeech 2026*

- **Shankar, N. B.**, Wang, Z., Eren, E., & Alwan, A. (2026). Compositional domain adaptation for automatic speech recognition with headwise selective attention merging. *Computer Speech & Language*
- Zhang, K., Shi, M., Eren, E., **Shankar, N. B.**, Wang, Z., & Alwan, A. (2026). STACodec: Semantic Token Assignment for Balancing Acoustic Fidelity and Semantic Information in Audio Codecs. *ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing*
- Wang, Zilai, **Shankar, N. B.**, Zhang, K., Wang, Zihan, & Alwan, A. (2026). Mind the Shift: Using Delta SSL Embeddings to Enhance Child ASR. *ICASSP 2026 – IEEE International Conference on Acoustics, Speech and Signal Processing*
- Bailey, A., Johnson, A., **Shankar, N. B.**, Veeramani, H., Washington, J., & Alwan, A. (2026). Addressing Bias in Spoken Language Systems Used in the Development and Implementation of Automated Child Language-Based Assessment. *Journal of Educational Measurement*, 63(1).
- Shi, M., **Shankar, N. B.**, Zhang, K., Wang, Z., & Alwan, A. (2025). Comparing Unsupervised and Supervised Semantic Speech Tokens: A Case Study of Child ASR. *AI for Children's Speech and Language Workshop (AI4CSL), ASRU 2025* (oral; **Best Paper Award**).
- **Shankar, N. B.**, Zhang, K., Mai, A., Shi, M., Long, A., Washington J., Morris, R., & Alwan, A. (2025). Leveraging ASR and LLMs for Automated Scoring and Feedback in Children's Spoken Language Assessments. *Proc. 10th Workshop on Speech and Language Technology in Education (SLaTE)*, 1–5.
- Zheng, J., Shetty, V., **Shankar, N. B.**, & Alwan, A. (2025). An Age-Agnostic System for Robust Speaker Verification. *Proc. 7th Workshop on Child Computer Interaction (WOCCI 2025)*
- Ying, A., **Shankar, N. B.**, Lin, C.-J., Shi, M., Wang, P., Shim, H. J., Arora, S., Van hamme, H., Alwan, A., & Watanabe, S. (2025). Benchmarking Training Paradigms, Dataset Composition, and Model Scaling for Child ASR in ESPnet. *Proc. 7th Workshop on Child Computer Interaction (WOCCI 2025)*, 6–10.
- **Shankar, N. B.**, Wang, Z., Zhang, K., Shi, M., & Alwan, A. (2025). CHSER: A Dataset and Case Study on Generative Speech Error Correction for Child ASR. *Proc. Interspeech 2025*, 2895–2899.
- **Shankar, N. B.**, Wang, Z., Eren, E., & Alwan, A. (2025). Selective Attention Merging for low resource tasks: A case study of Child ASR. *ICASSP 2025 - IEEE International Conference on Acoustics, Speech, and Signal Processing* (oral).
- **Shankar, N. B.**, Afshan, A., Johnson, A., Mahapatra, A., Martin, A., Ni, H., Park, H. W., Quintero Perez, M., Yeung, G., Bailey, A., Breazeal, C., & Alwan, A. (2024). The JIBO Kids Corpus: A speech dataset of child-robot interactions in a classroom environment. *JASA Express Lett.* 1 November 2024; 4 (11): 115201.
- Fan, R., **Shankar, N. B.**, & Alwan, A. (2024). Benchmarking Children's ASR with Supervised and Self-supervised Speech Foundation Models. *Proc. Interspeech 2024*, 5173–5177.
- Veeramani, H., Thapa, S., **Shankar, N. B.**, & Alwan, A. (2024). Large Language Model-based Pipeline for Item Difficulty and Response Time Estimation for Educational Assessments. *Proceedings of the 19th Workshop on Innovative Use of NLP for Building Educational Applications (BEA)*, 561–566.
- **Shankar, N. B.**, Fan, R., & Alwan, A. (2024). SOA: Reducing Domain Mismatch in SSL Pipeline by Speech Only Adaptation for Low Resource ASR. *2024 IEEE International Conference on Acoustics, Speech, and Signal Processing Workshops*, 560–564.
- Johnson, A., **Shankar, N. B.**, Ostendorf, M., & Alwan, A. (2024). An Exploratory Study on Dialect Density Estimation for Children and Adult's African American English. *The Journal of the Acoustical Society of America*, 155(4), 2836–2848.
- **Shankar, N. B.**, Johnson, A., Chance, C., Veeramani, H., & Alwan, A. (2024). CORAAL QA: A Dataset and Framework for Open Domain Spontaneous Speech Question Answering from Long Audio Files. *ICASSP 2024 - IEEE International Conference on Acoustics, Speech, and Signal Processing*, 13371–13375 (oral).

- Fan, R., **Shankar, N. B.**, & Alwan, A. (2024). UniEnc-CASSNAT: An Encoder-only Non-autoregressive ASR for Speech SSL Models. *IEEE Signal Processing Letters*, 31, 711–715.
- Johnson, A., Chance, C., Stiemke, K., Veeramani, H., **Shankar, N. B.**, & Alwan, A. (2023). An Analysis of Large Language Models for African American English Speaking Children’s Oral Language Assessment. *Journal of Black Excellence in Engineering, Science, and Technology*, 1.
- Johnson, A., Veeramani, H., **Shankar, N. B.**, & Alwan, A. (2023). An Equitable Framework for Automatically Assessing Children’s Oral Narrative Language Abilities. *Proc. Interspeech 2023*, 4608–4612.
- Veeramani, H., Johnson, A., **Shankar, N. B.**, & Alwan, A. (2023). Towards Automatically Assessing Children’s Oral Picture Description Tasks. *Proc. 9th Workshop on Speech and Language Technology in Education (SLaTE)*, 119–120.
- Palanisamy, G., **Shankar, N. B.**, Palanisamy, P., & Gopi, V. P. (2020). A hybrid feature preservation technique based on luminosity and edge based contrast enhancement in color fundus images. *Biocybernetics and Biomedical Engineering*, 40(2), 752–763.